HTTP(S) Responses

By now you've already had some exposure to the response side of HTTP. Many examples in this section have had curl running in verbose mode (thanks to the `-v` option), which will show both request and response headers. Let's just take another look at that output. Assuming you still have your `cli-server` container running, we'll do a simple fetch of `index.php` (the contents don't matter for our purposes here):

```
$ curl -v http://localhost:8080/index.php
...
* Request completely sent off
< HTTP/1.1 200 OK
< Host: localhost:8080
< Date: Mon, 14 Jul 2025 16:12:53 GMT
< Connection: close
< X-Powered-By: PHP/8.4.6
< Content-type: text/html; charset=UTF-8
...
* Closing connection
```

Everything after the `"* Request completely sent off"` line is the response. You'll notice the first line – `"HTTP/1.1 200 OK"` – looks a little like the first line of a request; this is the *status line*. A status line consists of the protocol version, a space, a *status code*, another space, then a "reason phrase" (which can be empty), terminated by a `CRLF` sequence. In this case our status code is `200`, and the reason phrase is `OK`; this tells the client that we didn't have any issues on our end and the content of the response should be considered correct (whether that's true or not is now a problem for the client and/or the end user). Status codes control fundamental behavior of the client; they might tell the client that they should visit a new location, or that their request wasn't made correctly, or even just that the server caught fire and died.

Following the status line, we have the response headers (one per line), and yes, they look pretty much the same as request headers. The header name and header value are delimited by a colon character, and each pair terminates with `CRLF`. You'll recognize `Content-Type:` with its IANA-approved media name (`"text/html"`). You should also note the use of an `X-Powered-By: PHP/8.4.6` "custom" header (despite such `X-` prefixing behavior being officially deprecated), which you'll see if using the PHP cli-server